

**DO NOT REMOVE FROM EXAM VENUE**

## **University of Plymouth**

**MODULE CODE: ELEC347 Examination**

**TITLE OF PAPER: Information & Communication  
Signal Processing**

**TIME ALLOWED** Three hours

**DATE** Tuesday 09 January 2024

**TIME** 09:00 – 12:00

**FACULTY** Science and Engineering

**SCHOOL** Engineering, Computing and Mathematics

**ACADEMIC YEAR** 2023/2024

**STAGE** Four

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### **INSTRUCTIONS TO CANDIDATES:**

The paper consists of **TWO** Sections.

**Section A:** Candidates should Attempt **ALL 10 QUESTIONS** in Section A.

**Section B:** Candidates should attempt any **TWO** questions from the five available in this section. All questions in this Section carry equal marks. Marks for parts of questions are shown where appropriate.

**Candidates are not permitted to look at the examination paper until instructed to do so.**

**SECTION A - Attempt ALL questions in this section.**

1. An analogue audio signal is sampled at the rate of 3.072 MHz with an oversampling ratio of 64. Determine the maximum frequency allowable in the analogue audio signal.  
[4 marks]
2. The signal at the second IF stage of a digital receiver has a bandwidth of 64 kHz. Assume the signal is centred at 3MHz; explain why the signal cannot be sampled at the minimum theoretical rate of 128 kHz using bandpass sampling. Justify your answer mathematically.  
[6 marks]
3. A real-time DSP system uses 8-bit ADC and operates at a sampling rate of 6.144 MHz. If the analogue input to the system is in the frequency range, 0-15 kHz, determine the effective resolution of the ADC.  
[6 marks]
4. In relation to DSP, explain the differences between Nyquist rate and Nyquist frequency.  
[4 marks]
5. A band pass FIR filter has a pass band that extends from 1000 to 1500 Hz, a transition width of 500 Hz and a sampling rate of 10 kHz. Determine the normalised band edge frequencies.  
[4 marks]
6. What is the necessary and sufficient condition for a realisable digital filter to have a linear phase characteristic?  
[4 marks]

7. A discrete-time filter has poles located at  $\pm 0.9j$  and zeros at  $\pm 1$ .
- (a) Sketch the pole-zero diagram and comment on the stability of the filter.  
[4 marks]
- (b) Sketch the magnitude frequency response of the filter. What type of filter is this (bandpass, band stop, low pass, high pass, and comb)?  
[3 marks]
8. State a key characteristic feature of the pole-zero diagram of a stable digital filter.  
[4 marks]
9. A 2-stage decimator is used to reduce the sample rate in a system from 240 k samples per second to 8 k samples per second. The decimation factors for the stages are  $M_1=15$  and  $M_2=2$ , and the filter lengths are  $N_1=45$  and  $N_2=43$ .
- Determine
- (a) The sample rates at the output of each stage.  
[2 marks]
- (b) The computational complexity for the system (in terms of multiplications and total storage).  
[4 marks]
10. In relation to radix-2 FFT flow graph
- (a) Sketch and label the butterfly diagram.  
[3 marks]
- (b) Write down the butterfly output equations.  
[2 marks]

**SECTION B - Attempt any TWO questions in this section.**

11. (a) State two main benefits of oversampling techniques in the design of analogue I/O interface for real-time DSP.

[2 marks]

- (b) The analogue front-end of a real-time DSP system is depicted in Figure Q11, together with its specifications in Table Q11a. Determine the following to meet the specifications, with the aid of appropriate sketches of the spectrum of the signal after sampling:

- (i) the cutoff frequency,  $f_c$ , for the anti-aliasing filter

[5 marks]

- (ii) the minimum sampling frequency,  $F_s$

Comment on the practical issues that should be taken into account in deciding the actual values for  $f_c$  and  $F_s$  to be used.

[8 marks]

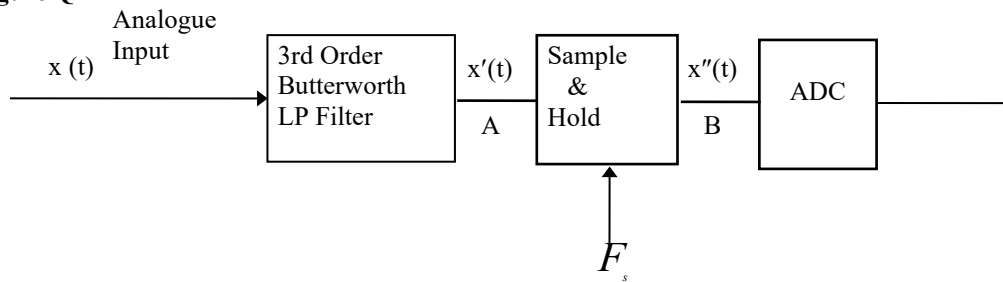
- (c) The DSP system in b) is to be used in a different application with the specifications for the analogue front-end given in Table Q11b. Explain how the system can be used, with the same analogue front-end, for the two applications using multirate techniques. What should the common  $f_c$  and  $F_s$  be? Justify your answer.

[10 marks]

**Question 11 continued ...**

## Question 11 continued

**Figure Q11**



**Table Q11a**

|                                                         |              |
|---------------------------------------------------------|--------------|
| Passband                                                | 0 - 4 kHz    |
| Maximum permissible passband ripple                     | 0.5 dB       |
| Ratio of signal to aliasing error level in the passband | $\geq 40$ dB |

Anti-aliasing filter characteristics:

$$H(f) = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^6}}$$

where  $f_c$  = 3 dB cutoff frequency of filter.

**Table Q11b**

|                                                         |              |
|---------------------------------------------------------|--------------|
| Passband                                                | 0 - 2 kHz    |
| Maximum permissible passband ripple                     | 0.5 dB       |
| Ratio of signal to aliasing error level in the passband | $\geq 40$ dB |

Anti-aliasing filter characteristics:

$$H(f) = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^6}}$$

where  $f_c$  = 3 dB cutoff frequency of filter.

12. (a) Explain, with the aid of appropriate sketches, the 'warping effect' associated with the Bilinear z-transform (BZT) design method for IIR filters. Comment on the practical implications of the 'warping effect' in applications where the response of the analogue filter to be digitised is close to the Nyquist frequency. Suggest a solution to the problem.

[5 marks]

- (b) A requirement exists for a digital lowpass filter with Butterworth characteristics and meeting the following specifications:

|                    |             |
|--------------------|-------------|
| Passband           | 0 – 3.4 kHz |
| Sampling frequency | 8 kHz       |
| Filter order       | 2           |

Starting with a suitable prototype analogue lowpass filter, determine the transfer function of the digital filter using the bilinear z-transform method. You may use the design information given below.

[10 marks]

- (c) (i) Explain, with the aid of suitable sketches, the four types of impulse response symmetry for linear phase FIR filters. What are the implications of the type of impulse symmetry on the group and phase delay characteristics of an FIR filter?
- (ii) A linear phase FIR filter has 75 coefficients and operates at a sampling rate of 10 kHz. Determine the delay through the filter.

[10 marks]

N.B: The transfer function of a second order prototype filter with Butterworth characteristics is given

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$$

**Question 12 continued ...**

### Question 12 continued

The lowpass-to-lowpass transformation is given by:

$$s \rightarrow \frac{s}{\omega'_p}$$

The bilinear z-transformation is given by:  $s \rightarrow \frac{z-1}{z+1}$

13. Figure Q13 (on the next page) shows the flowgraph for a 4-point, decimation in time, forward FFT.

- (a) Explain clearly how you would modify the flowgraph so that it can be used to perform inverse FFT. Justify your answer.

[6 marks]

- (b) Given the following DFT coefficients for a 4-point DFT:  $\{X(0) = 12, X(1) = -3-3j, X(2) = -2, X(3) = -3+3j\}$ . Use the modified flowgraph from (a) to obtain the corresponding discrete-time sequence (i.e.  $x(0)$ ,  $x(1)$ ,  $x(2)$  and  $x(3)$ ).

Assume that  $W^0 = 1$ ;  $W^1 = -j$  and that the butterfly equations are given by

$$A' = A + W^k B \text{ and } B' = A - W^k B \text{ (where the variables have the usual meaning).}$$

[12 marks]

- (c) Explain clearly how the FFT can be used to obtain the frequency response of each of the following types of digital filters:

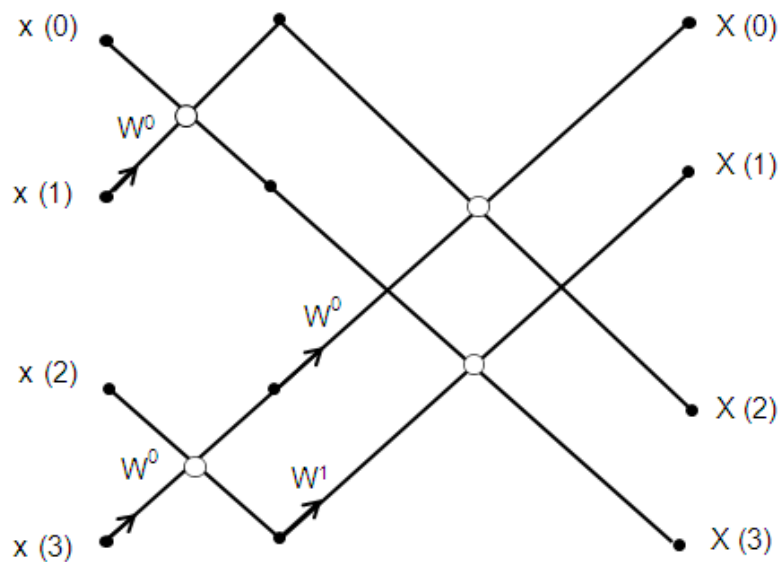
- (i) FIR filters
- (ii) IIR filters

What practical problems are associated with the above? Suggest how these may be overcome. Illustrate your answer with examples.

[7 marks]

**Question 13 continued ...**

**Question 13 continued**



**Figure Q13**

14. A third order IIR filter is characterised by the following transfer function:

$$H(z) = 0.8 \left( \frac{1 + 2z^{-1} + z^{-2}}{1 + 0.4z^{-2}} \right) \left( \frac{1 + z^{-1}}{1 + 0.6z^{-1}} \right)$$

- (a) Draw and label the realisation diagram for the filter. Assume a cascade realization using the direct form structure.

[8 marks]

- (b) Determine the difference equations for the filter, giving details of how you obtained your answer.

[8 marks]

- (c) Write a high-level language pseudo-code algorithm to implement the filtering operation. Explain how the algorithm works.

[9 marks]



15. (a) Explain how to change the sample rate of a multi-rate system by a non-rational factor. Illustrate your answer with an example.

[5 marks]

- (b) A requirement exists to reduce the sample rate of a system from 3 kHz to 2 kHz using multirate DSP techniques. Sketch and label the block diagram of a suitable sample rate converter to achieve this. Justify your answer.

[10 marks]

- (c) A 3-stage decimator down samples an audio signal from 3072 kHz to 48 kHz using decimation factors of 8, 4 and 2 for stages 1, 2, and 3 respectively. The band of interest is 0-20 kHz.

- (i) Sketch a block diagram of the decimator and indicate the sampling rate at the output of each stage.

[5 marks]

- (ii) Determine the band edge frequencies for the decimating filter at each stage. Sketch and label the decimating filters with the band edge frequencies.

[5 marks]

NB: The stop band for stage  $i$  is bounded by:

$(F_i - F_s / 2M) \leq f \leq F_{i-1} / 2, i = 1, 2, \dots, I$ , where  $F_i$  is the output sample rate at stage  $i$ ,  $F_s$  is the original sample rate and  $M$  is the overall decimation factor.